For the given matrix:

$$A = \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix}$$

Eigenvalues

```
F = eigen(A)
Λ = Diagonal(F.values)
```

where the Λ matrix is as follows:

$$\Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Julia's numerical output for the eigenvalues is:

$$\Lambda = \{-2.22045 \times 10^{-16} + 0i,\ 0 + 0i\}$$

The non-zero entry is on the order of 10^{-16} so it is likely some round off error that occured during numerical computation. When one hand computes the eigenvalues, we get:

$$\begin{aligned} \det(\lambda I - A) &= (\lambda - 1)(\lambda + 1) - i^2 \\ &= \lambda^2 + \lambda - \lambda - 1 + 1 \\ &= \lambda^2 = 0 \\ &\Rightarrow \lambda = 0 \end{aligned}$$

See that $\lambda = 0$ has algebraic multiplicity 2. We now check for the geometric multiplicity to confirm diagonalizability.

Eigenvectors

```
V = F.vectors
```

The set of eigenvectors v_1, v_2 both corresponding to $\lambda \approx 0$ is given as:

$$V = \{v_1,\ v_2\}, \qquad v_1 = \begin{bmatrix} -0.0 - 0.707107i \\ 0.707107 + 0.0i \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 0.707107 + 0.0i \\ 0.0 + 0.707107i \end{bmatrix}$$

Diagonalizability of A

For the geometric multiplicity, we find the rank of our eigenvector matrix V.

```
rank(V)
```

$$\text{rank}(V) = 1$$

So $\lambda = 0$ has geometric multiplicity 1, but algebraic multiplicity 2. Since these disagree, A is not diagonalizable. Additionally, A does not have an invertible V which satisfies $A = V\Lambda V^{-1}$.

This also follows directly and independently from $\Lambda = \text{diag}(0, 0)$ is the zero matrix. If A were diagonalizable, it would be similar to Λ, i.e. $A = S\Lambda S^{-1} = S \cdot 0 \cdot S^{-1} = 0$ for some invertible S, but a matrix similar to the zero matrix must *be* the zero matrix, and $A \neq 0$. Contradiction. So A is not similar to $\Lambda = 0$, and cannot be diagonalized.

Checking Normality of A

$$AA^* = \begin{bmatrix} 2 & -2i \\ 2i & 2 \end{bmatrix}, \qquad A^*A = \begin{bmatrix} 2 & 2i \\ -2i & 2 \end{bmatrix}, \qquad AA^* \neq A^*A$$

See that A is not normal, so by the Spectral Theorem A has no unitary eigenbasis i.e. $V^* \neq V^{-1}$. In our case, implementing $A = V\Lambda V^*$ asserting $V^* = V^{-1}$ was an error to begin with (recall $rank(V) = 1$, so V^{-1} cannot be computed either). Therefore, $C = V\Lambda V^*$ cannot equal A regardless of Λ. Now we numerically test if this conclusion is valid.

Testing $C = V\Lambda V^*$

```
C = V * Λ * V'
```

Julia's numerical output is:

$$C = \begin{bmatrix} -1.11022 \times 10^{-16} + 0.0i & 0.0 + 1.11022 \times 10^{-16}i \\ 0.0 - 1.11022 \times 10^{-16}i & -1.11022 \times 10^{-16} + 0.0i \end{bmatrix} \approx 0$$

Two reasons compound here: (1) A is non-normal, so $V^* \neq V^{-1}$ and this formula was never valid to begin with; (2) $\Lambda \approx 0$, so sandwiching a near-zero diagonal collapses the product to ≈ 0 regardless of V.

We can therefore conclude that the given matrix A is non-diagonalizable. Furthermore, unitary eigenbasis decompositions are also not possible due to the matrix's non-normality.